

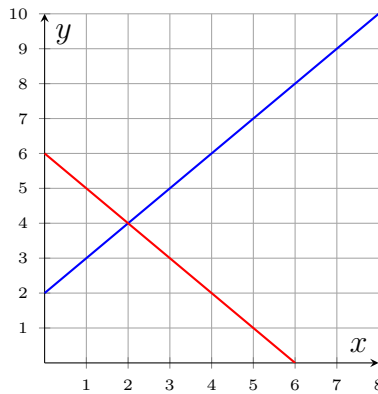
Solving Systems by Graphing

Graph both lines, read the crossing point, and check it in both equations

Directions

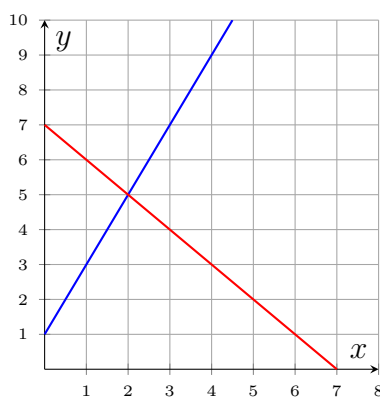
- Graph each line by plotting its y -intercept, then stepping by the slope. Draw both lines on the same grid.
- Read the coordinates of the crossing point, then substitute them into *both* equations to check. Use the space under each grid for that check.
- If the two lines never cross, say so instead of guessing a point.

1. Both lines are already drawn. Write the coordinates of the point where $y = x + 2$ and $y = 6 - x$ cross, then check the point in both equations.



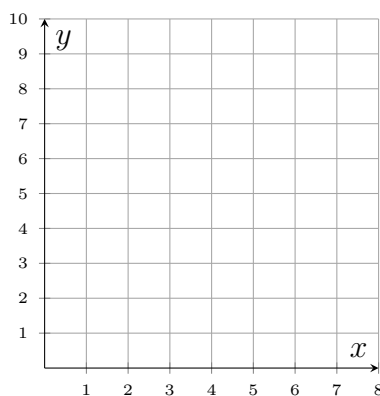
Answer: _____

- 2.** Both lines are already drawn. Write the coordinates of the point where $y = 2x + 1$ and $y = 7 - x$ cross, then check the point in both equations.



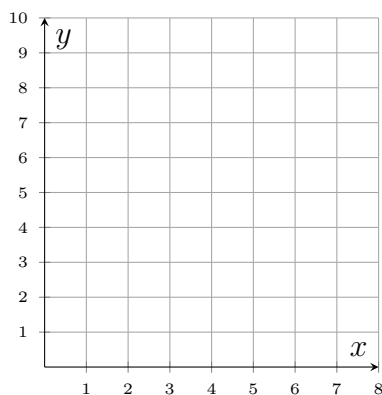
Answer: _____

- 3.** Graph $y = x + 1$ and $y = 10 - 2x$ on the grid, then write the coordinates of their intersection.



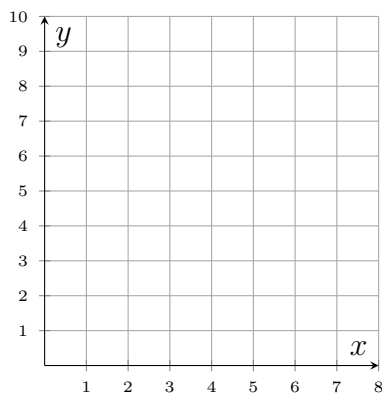
Answer: _____

4. Graph $y = 2x + 2$ and $y = x + 5$ on the grid, then write the coordinates of their intersection.



Answer: _____

5. Graph $y = 3x + 1$ and $y = 3x + 4$ on the grid.



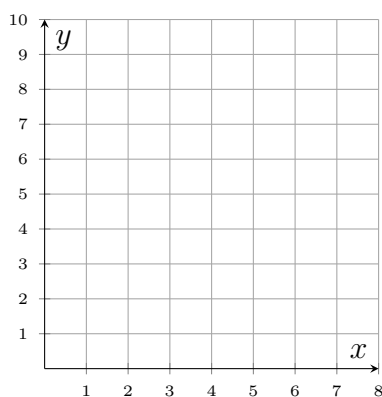
- (a) Write the solution of the system, or state that there is none.

Answer: _____

- (b) Explain what the two graphed lines show.

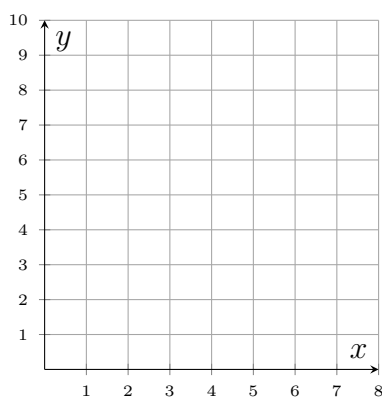
Answer: _____

- 6.** Graph $y = 8 - x$ and $y = \frac{1}{2}x + 2$ on the grid, then write the coordinates of their intersection.



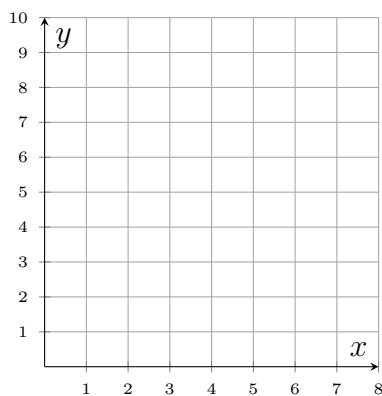
Answer: _____

- 7.** Graph $x + y = 8$ and $y = 3x$ on the grid, then write the coordinates of their intersection. (Find two points on the first line before you draw it.)



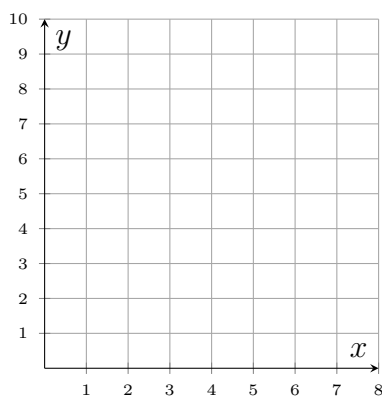
Answer: _____

8. Graph $x = 5$ and $y = 9 - x$ on the grid, then write the coordinates of their intersection.



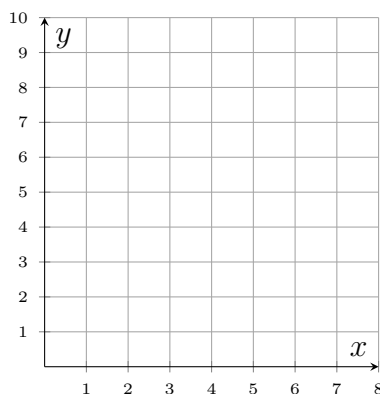
Answer: _____

9. A courier charges $y = 3x + 2$ dollars for a parcel of x kilograms; a rival charges $y = 5x$ dollars. Graph both on the grid and write the coordinates of the point where the two prices are equal.



Answer: _____

10. Consider the system $y = 2x + 3$ and $4x - 2y = -6$.



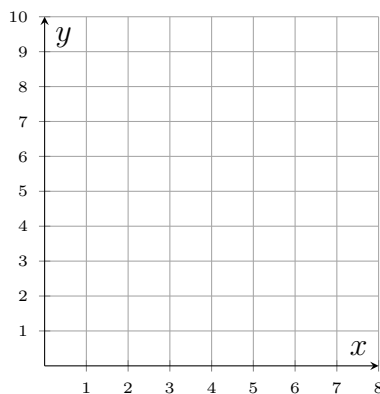
- (a) Solve the second equation for y , then graph both lines above.

Answer: _____

- (b) How many solutions does this system have? Explain why.

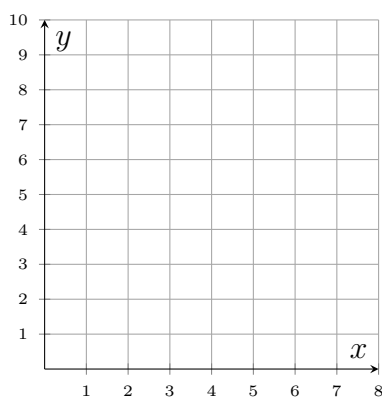
Answer: _____

11. Graph $2x + y = 12$ and $y = x + 3$ on the grid, then write the coordinates of their intersection.



Answer: _____

12. Company M charges $y = x + 6$ dollars for a trip of x miles; company N charges $y = 3x$ dollars.



(a) Graph both lines and write the coordinates of the point where the two prices are equal.

Answer: _____

(b) Explain which company is cheaper for a long trip, and how the graph shows it.

Answer: _____